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Quantitative analysis and visualization of chemical compositions during shrimp flesh deterioration using hyperspectral imaging: A comparative study of machine learning and deep learning models☆

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ABSTRACT

The current work explores hyperspectral imaging (HSI) to quantitatively identify changes in TVB-N and K value during shrimp flesh deterioration. The work developed low-level data fusion (LLF) and predictive models using both machine learning methods (PLS) and deep learning methods (CNN, LSTM, CNN-LSTM). Results indicate that deep learning methods show comparable performance due to their superior feature extraction and fitting capabilities, but traditional chemometric methods outperform deep learning models, achieving $R_p^2 = 0.9431$ (TVB-N), and $R_p^2 = 0.9815$ (K value). Subsequently, spatial distribution maps were generated based on the optimal predictive models to visualize the chemical composition changes in shrimp flesh. This approach allows for rapid, non-destructive prediction of spoilage-related changes. This technology can monitor shrimp quality in cold chain logistics, improve inventory management, and ensure seafood quality. Future research should optimize models for varied conditions and explore combining HSI method with other sensor technologies to enhance shrimp quality evaluation comprehensively and accurately.

1. Introduction

Shrimp flesh is rich in high-quality proteins, vitamins, and minerals, making it a highly favored seafood choice among consumers (Zhi et al., 2023). However, it is susceptible to spoilage during storage and transportation, leading to quality deterioration and food safety risks. Despite refrigeration extending its shelf life, microbial growth and enzymatic activity still contribute to quality degradation (Yu et al., 2018). Various factors influence shrimp flesh quality, including its nutritional composition, sensory attributes, freshness, tenderness, and water-holding capacity. Freshness is the most critical feature in assessing meat quality and the key indicators for spoilage include total volatile basic nitrogen (TVB-N) and lipid oxidation (K value) (Bekhit, Holman, Giteru, & Hopkins, 2021). Traditional methods for measuring indicators such as micro-diffusion, semi-micro nitrogen determination, Kjeldahl nitrogen, high-performance liquid chromatography (HPLC), polarography, and ion chromatography, are accurate but invasive and time-consuming. These approaches further suffer from lower efficiency and practical utility for fast and large-scale quality control. Given the limited shelf life of shrimp flesh, rapid and non-destructive detection technologies are imperative for assessing freshness, ensuring food safety, and minimizing economic losses.

Hyperspectral imaging (HSI) has emerged as a transformative tool for non-destructive food quality assessment, offering simultaneous spatial and spectral resolution to map chemical changes in real time (Özdoğan, Lin, & Sun, 2021). HSI enables concurrent capture of spectral and spatial data distributions from samples. This capability allows for the detection of microstructural and chemical changes in shrimp flesh during spoilage. While traditional spectroscopic techniques such as near infrared(NIR), Raman,focus on point measurements (Cheng et al., 2013), HSI excels in visualizing heterogeneous degradation patterns in complex matrices like shrimp flesh. Recent applications span vegetables (Yu et al., 2023), grains (Guo et al., 2023), and meats (Jo et al., 2024), yet its potential for dynamic monitoring of shrimp flesh spoilage remains

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underexplored. Critically, existing HSI studies on food quality assessment have predominantly relied on shallow machine learning (ML) models—such as partial least squares (PLS), support vector machine (SVM), and artificial neural networks (ANN) (Li et al., 2020; Nicolaï et al., 2007), neglecting the temporal dimension of spoilage and the synergy between multimodal data fusion and deep feature learning—a gap this study aims to bridge. This study aims to address these gaps by introducing two key innovations: deep learning algorithms and a spectral fusion strategy. Deep learning algorithms have gained significant attention across various fields due to their advanced feature learning capabilities, making them a promising alternative for improving model performance in food quality and safety assessment (Ouyang, Fan, Chang, Shoaib, & Chen, 2025). Moreover, fusing visible and near-infrared information provides more comprehensive insights and a deeper understanding of results compared to individual spectral blocks alone (Cheng, Sun, Shi, & Dai, 2024).

The combined use of deep learning and data fusion strategies for food quality remains in its early stages of exploration (Saha & Manickavasagan, 2021), making it a frontier worthy of deeper investigation. The current work explores HSI, ML, deep learning, and data fusion methods to estimate TVB-N and K value during the spoilage process of shrimp flesh. The study primarily involves the following aspects: 1) acquisition of hyperspectral images and measurement of TVB-N and K value during shrimp flesh spoilage; 2) ML of single and fused data using traditional methods; 3) application of deep learning methods to model single and fused data; 4) selection of the optimal model for visualizing content distribution. The schematics of the work in Fig. 1 briefly illustrate the process. This study comparatively analyzes ML approaches with deep learning methods in the context of hyperspectral modeling, providing a theoretical basis and guidance for industrial applications and modeling methodologies.

2. Materials and methods

2.1. Shrimp flesh samples preparation

Fresh, undamaged shrimp (*Litopenaeus vannamei*) samples were obtained from the Xiamen Xishang International Aquatic Products Trading Center and transported to the laboratory in plastic bags oxygenated with oxygen. Upon arrival at the laboratory, the shrimp were rapidly chilled using crushed ice to lower their body temperature to below 0 °C, inducing death. The shrimp were then peeled, deheaded, and eviscerated. A set of 150 shrimp flesh samples, each weighing around 20 g with an allowable variance of ±2 g, was collected and individually sealed in

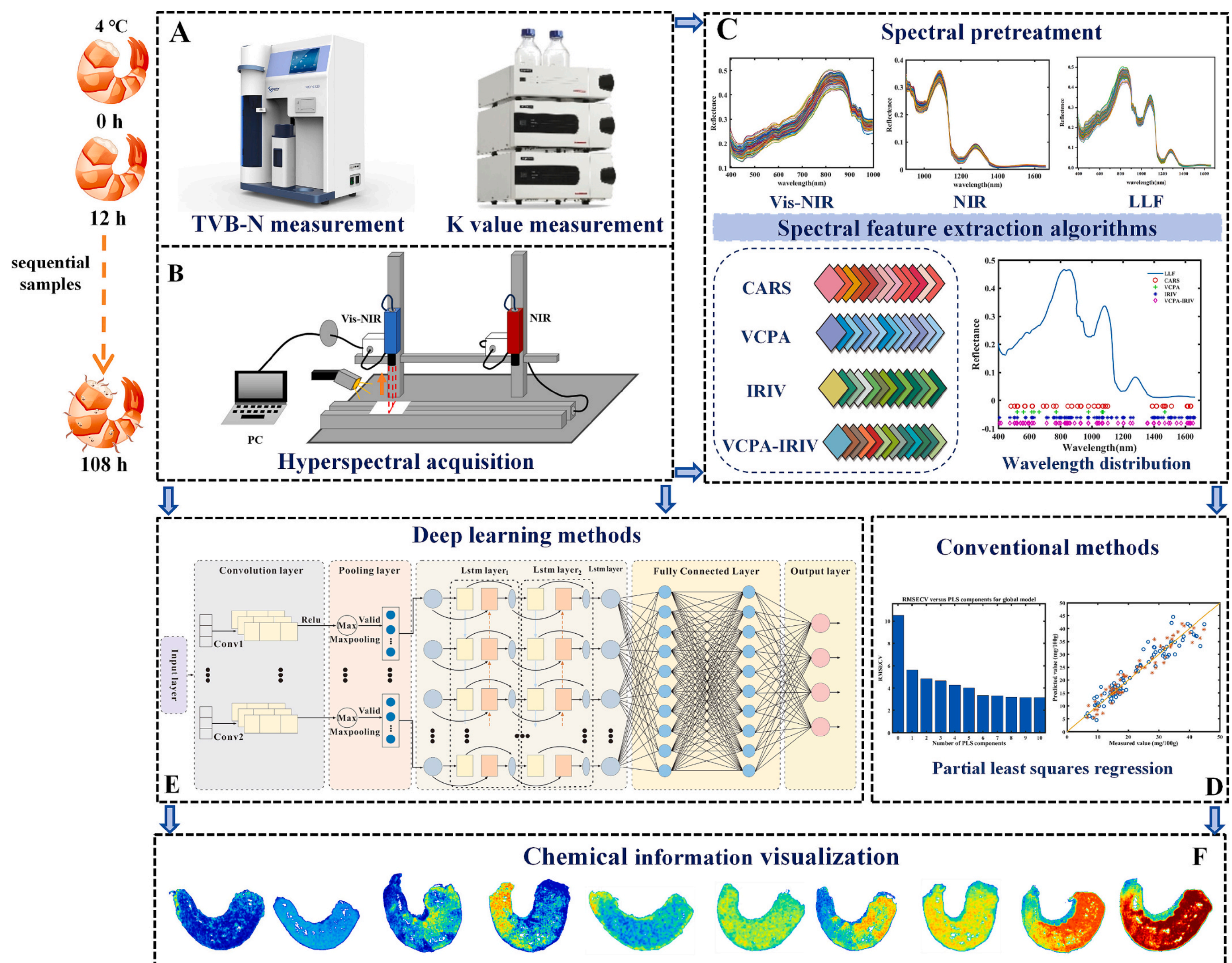

Fig. 1. Schematic diagram of the whole research process. (A) Measurement of TVB-N and K value content in shrimp flesh; (B) Hyperspectral acquisition; (C) Spectral pretreatment and characteristic wavelength selection; (D) Development of machine learning methods; (E) Development of deep learning methods. (F) Chemical information visualization.

sterile bags for storage at 4 °C. From the start of the experiment (0 h), 15 samples were randomly selected every 12 h, and sampling continued until the experiment concluded, lasting a total of 108 h. Spectral data from the samples were collected using the HSI system, covering a wavelength range from visible-near infrared (Vis-NIR, 400–1000 nm) to NIR (900–1700 nm). To correlate the spectral data with actual quality indicators, traditional chemical analyses were conducted simultaneously to measure the content of TVB-N and K value, which are indicators of spoilage and freshness. Chemical analyses and spectral data collection were conducted concurrently, with samples taken every 12 h.

2.2. Reference measurement

Detailed information about all analytical grade reagents has been incorporated into the supporting materials (Table S1).

2.2.1. TVB-N measurement

The TVB-N content in shrimp flesh was assessed via the automated Kjeldahl method, as per the Chinese national standard GB 5009.228–2016 method procedure. The measurement was performed with the Haineng (Shandong, China) K9860 fully automatic Kjeldahl nitrogen analyzer. Specifically, 10 g of homogenized shrimp flesh was combined with 75 mL of water in a distillation flask and soaked for half an hour. Following this, an accurate weight (1 g) of magnesium oxide (MgO) was introduced into the distillation tube and promptly connected to the analyzer for analysis.

2.2.2. K value measurement

The K value content in shrimp flesh was examined with HPLC following the SC/T 3048–2014 standard of the aquaculture industry, in China. Initially, 10 % perchloric acid was added to the homogenized shrimp meat sample. After vortex oscillation for one minute, the mixture was centrifuged in a frozen centrifuge for 10 min to obtain the supernatant. This step was repeated by adding 10 mL of 5 % perchloric acid and centrifuging for another 10 min. The supernatants were pooled and the pH was adjusted to 6.0–6.4 using a sodium hydroxide solution. The combined supernatant was diluted to 50 mL with chilled water (4 °C), filtered, and subsequently analyzed using the Tongwei high-performance liquid chromatography instrument, EasySep®-3030. The K value was then computed using the equation below.

$$K = \frac{\mathrm{HxR} + \mathrm{Hx}}{\mathrm{ADP} + \mathrm{AMP} + \mathrm{IMP} + \mathrm{HxR} + \mathrm{Hx}} \times 100\% \tag{1}$$

where *HxR* presents Hypoxanthine Riboside; *Hx* denotes Hypoxanthine; *ADP* stands for Adenosine Diphosphate; AMP indicates Adenosine Monophosphate; IMP corresponds to Inosine Monophosphate.

2.3. Acquisition and extraction of hyperspectral images

2.3.1. Hyperspectral image acquisition

The instruments used in this study were a push-sweep HSI system manufactured by Wuling Optics (Shanghai, China). Each HSI system comprises a spectral line-scan imaging spectrometer, a CCD camera, a camera lens, and two tungsten lamps (LS-150, Wuling Optics, Taiwan). Specifically, the Vis-NIR system features a line-scan imaging spectrometer with a 2.8 nm resolution (V10E, Specim, Finland), coupled with a CCD camera offering a resolution of 1600 × 1100. In contrast, the NIR system incorporates an imaging spectrometer with 4 nm resolution (N17E, Specim, Finland) and a camera boasting a resolution of 640 × 512. Both systems rely on a stepper motor-driven moving platform to facilitate the acquisition of hyperspectral images through the instrument software.

2.3.2. Spectra extraction

After collecting hyperspectral image data, correcting the original image and extracting the region of interest (ROI) before model establishment is essential. This correction involves using white and dark reference images to mitigate spectral distortion caused by light source, instrument response and environmental factors. Both Vis-NIR and NIR hyperspectral images of all shrimp flesh were calibrated using these reference images according to formula (2):

$$I = \frac{I_0 - R_b}{R_w - R_b} \times 100 \tag{2}$$

where R_b and R_w refer to the black and white reference images, respectively. I_0 denotes the original hyperspectral image, while I represents the corrected image.

Spectral characteristics of shrimp flesh ROI were extracted from HSI data using a thresholding method, reducing subjectivity compared to manual ROI selection in ENVI. Bands with the lowest reflectance for the sample and highest for the background were selected, and the thresholds of Vis-NIR and NIR were at the 250th band (0.39) and the 287th band (0.054), respectively. A binary image was created after thresholding, and ROI locations were determined for spectral extraction. Reflectance thresholds at specific wavelengths were applied to define shrimp flesh spectral signatures, with average values computed as spectral features. This method ensures objective and accurate data for spectral analysis and model development.

2.4. Data analysis methods of machine learning

2.4.1. Spectral pretreatment

The HSI images collected are liable to be affected by factors originating from the sensor system itself as well as external environmental conditions. This results in noise, varying degrees of radiation distortion, and geometric distortions. Preprocessing hyperspectral data can significantly improve the signal-to-noise ratio (SNR), geometric accuracy, and radiometric accuracy. Several pretreatment methods, including standard normal variate (SNV), multivariate scattering correction (MSC), first derivative (1stDer), second derivative (2ndDer) and Savitzky-Golay (SG) smoothing filtering were tested. The optimal preprocessing technique was chosen based on the evaluation metrics.

2.4.2. Spectral feature information extraction

The original Vis-NIR and NIR datasets respectively contain 678 and 465 wavelengths. Numerous features and unrelated variables can increase model operation time and impact its performance. Feature optimization algorithms have the capability to decrease feature count and enhance model efficiency (Ouyang et al., 2023). This study employed competitive adaptive reweighted sampling (CARS) (Li, Liang, Xu, & Cao, 2009), variable combination population analysis (VCPA) (Yun et al., 2015), iterative retention of informative variables (IRIV) (Yun et al., 2014), and iterative reserved information variable (VCPA-IRIV) (Yun et al., 2019) for spectral feature extraction, data fusion and subsequent modeling purposes.

2.4.3. PLS model

The PLS model is a statistical method ideal for establishing predictive models in scenarios with high collinearity or low sample size. By minimizing the residuals between observed variables and model predictions, the method optimizes the model and successfully integrates the benefits of principal component analysis (PCA) with ordinary least squares (OLS). This approach effectively establishes models relating spectral data to chemical information.

2.5. Deep learning methods

2.5.1. CNN

The deep learning convolutional neural network (CNN) structure is inspired from biological visual processing mechanisms, fundamentally transforming ML. In this study, one-dimensional CNN (1D-CNN) regression model was tailored for processing one-dimensional hyperspectral data. The specific structure and parameters are shown in Table S2. The model comprises 2 convolutional layers, 2 batch normalization layers, 2 max-pooling layers, 1 flattening layer, 1 dropout layer, 1 fully connected layer, and 1 output layer. The model adeptly captures spatial features within the input spectrum by iteratively employing convolutional layers. Rectified Linear Units (ReLU) are utilized as activation functions across all convolutional layers, with a pooling size set at 4 × 1 and a stride of 2.

2.5.2. LSTM

Long Short-Term Memory (LSTM) is a recurrent neural network distinguished by its foundational architecture, which consists of three gates: the input gate (i_t), forget gate(f_t), and output gate (o_t). along with a cell state (C_t). The internal workings can be summarized by the following equations:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{3}$$

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{4}$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \tag{5}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{6}$$

$$h_t = o_t \cdot tanh(C_t) \tag{7}$$

where x_t represents the input data and h_{t-1} refers to a previous hidden state, W_i, W_f, W_C, W_o and b_i, b_f, b_C, b_o represent the weight and bias parameters, respectively.

LSTM harnesses specialized gating mechanisms to precisely manage information flow, thereby enhancing the network's capacity to learn and retain long-term dependencies. This capability gives LSTM significant advantages in handling sequential data tasks (Geng et al., 2023). The schematics of the LSTM can be seen in Figure S1.

2.5.3. CNN-LSTM

For CNN and LSTM, modeling results for the same dataset can differ due to their distinct strategies and perspectives in handling problems. Each model possesses unique strengths and limitations, motivating the construction of a CNN-LSTM regression model. Figure S2 depicts this structure, the CNN layer initially learns local features from hyperspectral data through convolutions, pooling, and other operations, transforming them into a temporal feature sequence suitable for LSTM processing. Subsequently, these feature sequences are passed to the LSTM layer for further refinement. The LSTM layer effectively utilizes the temporal information within these feature sequences to establish long-term dependencies among spectral data. As compared to their use alone, the combined use of CNN-LSTM comprehensively captures both spatial and temporal dependencies in the data, thereby enhancing predictive capability (Qin et al., 2024).

2.6. Data fusion strategy

A low-level data fusion (LLF) technique was used to integrate information from diverse sources, leveraging their unique characteristics to extract richer insights than possible with a single data type. Specifically, the raw spectral data from Vis-NIR (649 variables) and NIR (427 variables) were used. To ensure data reliability, sample sizes from each source were kept identical during merging.

2.7. Model evaluation

In this modeling procedure, 70 % of the dataset was designated for calibration based on sample set partitioning using joint X-Y distances (SPXY), whereas the remaining 30 % was allocated for prediction. The model performances were assessed based on the coefficient of determination for the training set (R_c^2), the coefficient of determination for the prediction set (R_p^2), the root mean square error of calibration (RMSEC), the root mean square error of prediction (RMSEP), the mean absolute error of calibration (MAEc), and the mean absolute error of prediction (MAEp). An effective model should possess a higher R_c^2 and R_p^2, and lower RMSEC, RMSEP, MAEc, and MAEp (Guo et al., 2020). The residual predictive deviation (RPD) was also used, with an RPD greater than 2 indicating good performance. The evaluation parameters are as follows:

$$R_c^2, R_p^2 = 1 - \frac{\sum_{i=1}^{n} (\widehat{y}_i - y_i)^2}{\sum_{i=1}^{n} (y_i - y_m)^2} \tag{8}$$

$$RMSEC, RMSEP = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (\widehat{y}_i - y_i)^2} \tag{9}$$

$$MAEc, MAEp = \frac{1}{N} \sum_{i=1}^{N} |\widehat{y}_i - y_i| \tag{10}$$

$$RPD = \frac{1}{\sqrt{1 - R_p^2}} \tag{11}$$

where n is the sample size of the calibrated or predicted subset; y_i and $\widehat{y}_i$ represent measured value and predicted value i_{th} sample; y_m is the average of the measurements for a calibrated or predicted subset.

2.8. Chemical information visualization

HSI data is a three-dimensional data cube containing spectral information at multiple wavelengths, making it suitable for material composition analysis and distribution visualization. In this study, the hyperspectral data visualization process begins by converting the spectral image into a 2D matrix of size MN × K, where M and N represent the spatial dimensions and K is the number of wavelengths. A model is then built for each chemical component (TVB-N and K value) to predict the chemical composition at each pixel. The model is applied to the two-dimensional matrix to calculate the chemical composition for each pixel, which is subsequently normalized to the grayscale range [0, 255] to generate grayscale images. A median filter with a five-pixel neighborhood is used to reduce noise while preserving structural information. Finally, pseudocolor processing is applied to enhance image visibility, making the component distribution more intuitive. The concentration values are displayed using a color bar. This approach improves the visualization of hyperspectral images and provides both quantitative information about components and their spatial distribution.

2.9. Software

This study utilized PyCharm Community Edition 2020 (with Python 3.9) for extracting and visualizing spectral data, while data processing and model construction were completed in Matlab R2022a. All data analysis processes were conducted on a computer equipped with an Intel i9–13,900 KF CPU, 32 GB RAM, and an NVIDIA GeForce RTX 4090 GPU. The plotting software used was Origin 2021. The libraries employed include NumPy 1.26.3, Pandas 2.1.4, Spectral 0.23.1, and others. The selection of these tools and platforms ensures the efficiency and accuracy of data processing.

3. Results and discussion

3.1. Reference chemical compositions and spectral analysis

As illustrated in Fig. 2A, the concentrations of TVB-N and K value in shrimp flesh samples exhibited a progressive increase over time during

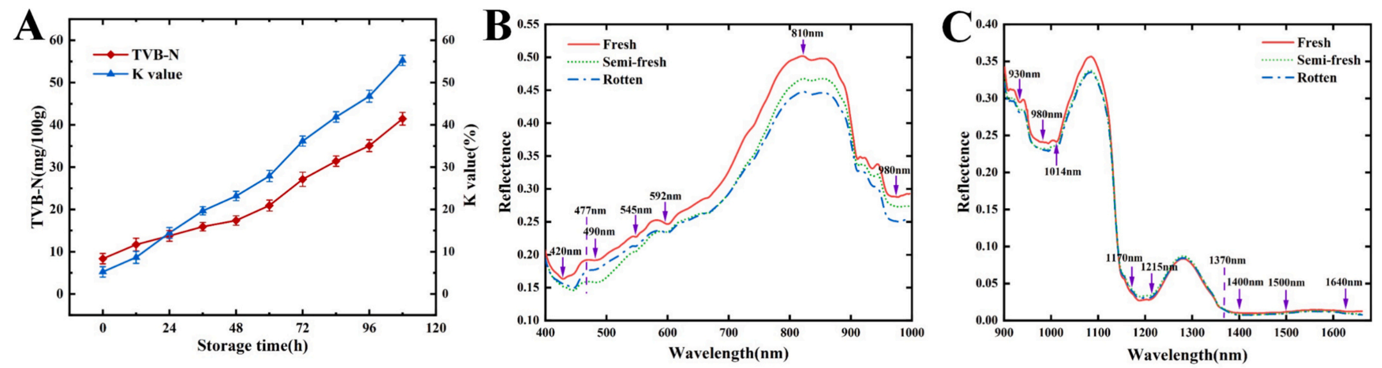

Fig. 2. (A) Change trend of TVB-N and K value; (B) Average Vis-NIR with different degrees of freshness; (C) Average NIR with different degrees of freshnes(Tang et al., 2024)s;

the spoilage process. Initially, the average TVB-N and K value were 8.36 mg/100 g and 5.24 % at 0 h, respectively, similar to previous reports (Qian et al., 2015). Generally, shrimp flesh with TVB-N less than 15 mg/100 g and K value less than 20 % is considered fresh; shrimp flesh with TVB-N between 15 and 30 mg/100 g and K value between 20 % and 40 % is considered semi-fresh; while shrimp flesh with TVB-N greater than 30 mg/100 g and K value greater than 40 % indicates spoilage and it is no longer suitable for human consumption. Therefore, in this study, the shrimp flesh reached a semi-fresh state at 36 h and became spoiled and unsuitable for human consumption at 84 h. This spoilage trend is similar to the report investigated by Dai et al. (Dai, Cheng, Sun, Zhu, & Pu, 2016).

3.2. *Spectral analysis*

Fig. 2 illustrates average spectral profiles of shrimp flesh at different freshness levels during storage at 4 °C. Fig. 2B represents the Vis-NIR spectrum, while Fig. 2C shows the NIR spectrum. The intensity of spectral reflectance for both spectra decreases as shrimp flesh deteriorates, which corresponds to an increase in chemical markers of spoilage. This trend suggests a negative relationship between spectral reflectance and shrimp freshness. The reflectance of fresh samples exhibits a larger difference with semi-fresh and rotten samples, while reflectance between semi-fresh and rotten samples is smaller.

In Vis-NIR, absorption peaks are observed at 420, 477, 490, 516, 545, 592, 710, 783, and 980 nm. Studies indicate that the absorption peak at 420 nm is attributed to Soret absorption, whereas peaks at 477, 490, 516, 545, and 592 nm are linked to pigment absorption, specifically hemoglobin and myoglobin. Additionally, absorption peaks around 760 and 980 nm are associated with the third and second overtones of O—H stretching in water (Li et al., 2024; Pu, Sun, Ma, & Cheng, 2015). The high reflectance observed at 810 nm could result from the combined influence of -CH stretch overtones within protein content and -OH stretch overtones in water (Dai et al., 2016).

In NIR, the band at 930 nm is usually associated with fat-associated -CH stretching overtones (Baek, Lee, Mo, Chan, & Kim, 2021). The absorption peaks at 980 and 1014 nm are indicative of the second overtone of -OH stretching within water, and the strong absorption peaks around 1170–1215 nm are ascribed to -CH, the second overtone of -CH_2 and -CH_3 stretching in fat (Li et al., 2023). Broad absorption regions of 1400 to 1500 nm are associated with the first overtone combination stretching of -CH and -CH_2 in fat, and the first overtone stretch absorption of -NH in proteins between 1450 and 1500 nm. In addition, these two less pronounced absorption zones in fat samples near 1370 nm and 1640 nm are assigned to the first overtone of -CH_3 stretching fat (Yu, Wang, Wen, Yang, & Zhang, 2019).

3.3. *Results of machine learning*

3.3.1. *Result of spectra preprocessing method*

Spectral data frequently encounters instrument noise and stray light challenges, which can compromise the precision and dependability of the resultant model. Consequently, preprocessing of the original spectrum is typically essential prior to modeling. Table 1 shows how different pretreatment methods affect the model's predictive efficiency. Preprocessing significantly enhances model performance when compared to using raw data. Notably, Vis-NIR, NIR, and LLF data consistently selected the 2ndDer as the optimal preprocessing method. Based on the second derivative preprocessed TVB-N prediction model, the results showed an R_p^2 of 0.8181 and an RPD of 2.36. For the K value prediction model, the R_p^2 was 0.8059 with an RPD of 2.29, whereas for the LLF prediction, the R_p^2 was 0.8297 with an RPD of 2.44. This suggests that the 2ndDer enhances the accuracy of the model's prediction. Generally, this improvement can be attributed to the 2ndDer's ability to enhance resolution, remove background or noise, amplify differences between spectra, and enhance the precision and reliability of peak detection (Genkawa, Ahamed, Noguchi, Takigawa, & Ozaki, 2016).

3.3.2. *Results of variable selection based on the PLS*

The predictive results of the PLS quantitative models established using CARS, VCPA, IRIV, and VCPA-IRIV feature variable extraction algorithms are presented in Table 2. Each model is implemented for 20 runs, and the average value of the results is computed to reduce the randomness, mitigate the risk of overfitting and enhance the robustness of the model. The PLS models established through variable selection algorithms have consistently demonstrated excellent prediction performance, with R_p^2 values above 0.85 and RPD values exceeding than 2 in all cases. This indicates that the models possess robust predictive capabilities. For the optimal models predicting each chemical component, the R_p^2 values are all above 0.9 and the RPD values exceed 3, suggesting that the models are capable of highly reliable predictions under various conditions. Specifically, in the prediction of TVB-N, the IRIV based on LLF data exhibits superior performance (R_p^2 = 0.9431, RMSEP = 2.49 mg/100 g, and RPD = 4.23). In the prediction of K value, VCPA-IRIV based on LLF data outperforms other models to obtain the best prediction effect, with R_p^2 of 0.9815 and RPD of 7.40. The LLF data provides superior performance compared to individual Vis-NIR and NIR data blocks. This advantage arises from the synergistic interaction between Vis-NIR and NIR spectra. Vis-NIR is sensitive to the oxidation state transitions of myoglobin on the shrimp surface, which can trace the self-dissolution enzyme release process associated with TVB-N (Zhang et al., 2022). NIR primarily responds to C-H/N-H bond vibrations (1200–1700 nm) and captures aldehyde and ketone compounds generated from lipid oxidation (Grebenteuch, Kanzler, Klaußnitzer, Kroh, & Rohn, 2021). By

Table 1
Comparison of PLS models based on different pre-processing methods of Vis-NIR, NIR and LLF data.

Data	Methods	LVs	VARs	Calibration			Prediction			RPD
				R^2c	RMSEC	MAEc	R^2p	RMSEP	MAEp	
Vis-NIR	RAW	10	678	0.8353	4.22	3.25	0.7029	5.69	4.22	1.85
	SNV	10		0.8911	3.43	2.52	0.7097	5.62	4.09	1.87
	MSC	10		0.8892	3.46	2.55	0.7402	5.32	3.93	1.98
	1stDer	8		0.8974	3.33	2.43	0.7227	5.49	3.97	1.92
	2ndDer	**7**		**0.9239**	**2.87**	**2.20**	**0.8181**	**4.45**	**3.58**	**2.36**
	SG	10		0.8343	4.24	3.26	0.7021	5.70	4.23	1.85
NIR	RAW	10	465	0.8537	4.12	3.22	0.7982	4.89	3.73	2.22
	SNV	10		0.8545	4.11	3.19	0.7948	4.87	3.91	2.23
	MSC	10		0.8528	4.13	3.21	0.7905	4.95	3.88	2.20
	1stDer	8		0.8923	3.54	2.74	0.8020	4.78	3.55	2.27
	2ndDer	**7**		**0.9164**	**3.11**	**2.36**	**0.8059**	**4.73**	**3.75**	**2.29**
	SG	10		0.8271	4.48	3.51	0.7833	4.99	4.06	2.17
LLF	RAW	10	1079	0.8325	4.26	3.37	0.7153	5.57	4.10	1.89
	SNV	9		0.8298	4.29	3.38	0.6717	5.98	4.26	1.76
	MSC	9		0.8267	4.33	3.42	0.6828	5.88	4.25	1.79
	1stDer	8		0.8971	3.34	2.47	0.7890	4.79	3.69	2.20
	2ndDer	**6**		**0.8959**	**3.36**	**2.64**	**0.8297**	**4.31**	**3.37**	**2.44**
	SG	10		0.8314	4.27	3.38	0.6826	5.59	4.11	1.88

Notes: LVs, the latent variables.
VARs, The number of selected wavelength variables.
R^2C, Coefficient of determination of calibration.
R^2P, Coefficient of determination of prediction.
RMSEC, Root mean square error of calibration.
RMSEP, Root mean square error of prediction.
MAEc, Mean absolute error of calibration.
MAEp, Mean absolute error of prediction.
RPD, Residual predictive deviation of prediction.

integrating latent variables (LLF), the model effectively extracts the synergistic responses of spoilage markers across spectral ranges, which is consistent with the findings of Tang et al. (Tang et al., 2024). In previous studies, mid-infrared technique (Zhou et al., 2023) and line-scan spatial offset Raman spectroscopy (Liu, Huang, Zhu, Qin, & Kim, 2021) have been applied in the quality assessment of shrimp. These methods involve relatively simpler data processing steps as compared to hyperspectral technology. However, the spatiotemporal resolution advantage of HSI is significantly superior to that of line-scan spatial offset Raman spectroscopy, owing to its contactless optical design and parallel spectral acquisition architecture. Additionally, unlike the mid-infrared technique, which relies on the interferometric principle of Fourier transform, the scanning imaging of HSI is better suited for dynamic detection scenarios in production lines.

3.3.3. Wavelength distribution based on optimal model

Utilizing various optimization algorithms, precise feature wavelengths were selected across datasets to assess and improve the predictive capabilities of the data fusion techniques. As shown in Fig. 3, for the TVB-N prediction, the number of effective feature wavelength variables chosen by LLF is fewer than the sum obtained by individually using Vis-NIR and NIR screening variables. LLF primarily optimized the spectral bands selected between 1370 and 1640 nm, which correspond to prominent absorption features associated with -NH stretching and -CH_3 stretching, both of which are related to fat content. These stretching vibrations are sensitive to changes in fat composition, which can influence the spoilage rate of shrimp flesh. The advantage of LLF becomes evident when compared to using Vis-NIR and NIR data alone. LLF screened key absorption peaks at 420, 477, 490, 545, and 592 nm in the Vis-NIR spectrum for predicting the K value, which likely correlates with changes in muscle pigment and texture during refrigeration. These changes can induce vibrations in adenosine triphosphate (ATP)-related chemical bonds, as described by Cheng et al. (Cheng, Sun, Pu, & Zhu, 2015). Moreover, this improvement in prediction stems from the ability of LLF to discern and prioritize spectral features that reflect both the chemical and physical modifications occurring in shrimp muscle during refrigeration.

3.4. Results of deep learning methods

A one-dimensional deep learning model was constructed based on CNN, LSTM and CNN-LSTM. Vis-NIR, NIR and LLF spectral data were used to determine the TVB-N and K value generated during shrimp flesh putrefaction at the same time. The “MSE” loss function and “Relu” activation function were employed across all three models for all data types. The prediction results shown in Table 3 represent the averages of 20 runs. All three models exhibited satisfactory predictive capabilities, with LSTM generally outperforming CNN. The possible reason why LSTM is superior to 1D-CNN is consistent with the time-resolved characteristics of shrimp metamorphic biochemistry. During cold storage, ATP degradation (K value) follows a sequential pathway (ATP → ADP → AMP → IMP→Hx), while the accumulation of TVB-N is driven by progressive proteolysis (Cheng et al., 2015). LSTM's gated cycle unit explicitly simulates these time dependencies, capturing phase transitions such as the critical transition from aerobic to anaerobic metabolism 48 h after death. In contrast, the translation invariance of 1D-CNN ignores such sequence-sensitive spectral dynamics, despite its dexterity in local feature extraction. Compared to individual CNN and LSTM models, CNN-LSTM exhibited superior predictive performance, with R_p^2 improvements of approximately 0.3 to 0.5 and RPD enhancements of 0.2 to 2.2 for each data set. This is because the CNN-LSTM model realizes cross-scale feature fusion through cascading architecture, effectively utilizing the advantages of the two algorithms. It is noteworthy that although the deep learning model still exhibits the best performance on the LLF data, its performance remains inferior to that of the machine learning methods. The possible reason is that hyperspectral data typically contains hundreds of spectral bands, but the sample size is

Table 2
Comparison of PLS models based on different feature extraction algorithms of Vis-NIR, NIR and LLF data.

Data	Compositions	Methods	LVs	VARs	Calibration			Prediction			RPD
					R^2c	RMSEC	MAEc	R^2p	RMSEP	MAEp	
Vis-NIR	TVB-N	CARS	4	27	0.8988	3.31	2.57	0.8630	3.86	3.09	2.72
		VCPA	9	11	0.9160	3.02	2.35	0.8899	3.46	2.70	3.31
		IRIV	6	53	0.9443	2.466	1.83	0.8980	3.33	2.59	3.16
		VCPA-IRIV	**11**	**69**	**0.9588**	**2.11**	**1.61**	**0.9233**	**2.89**	**2.27**	**3.64**
	K value	CARS	13	19	0.9629	2.74	2.04	0.9173	4.09	2.93	3.51
		VCPA	10	13	0.9649	2.66	2.02	0.9493	3.19	2.283	4.48
		IRIV	**13**	**24**	**0.9700**	**2.46**	**1.82**	**0.9552**	**3.01**	**2.13**	**4.76**
		VCPA-IRIV	11	20	0.9823	2.66	1.99	0.9485	3.23	2.31	4.44
NIR	TVB-N	CARS	10	23	0.8797	3.74	2.86	0.8591	4.03	3.06	2.69
		VCPA	9	13	0.8832	3.68	2.82	0.8812	3.70	2.92	2.93
		IRIV	**10**	**31**	**0.9191**	**3.25**	**2.33**	**0.9039**	**3.33**	**2.62**	**3.25**
		VCPA-IRIV	11	42	0.9176	3.09	2.11	0.9016	3.37	2.58	3.21
	K value	CARS	10	23	0.9629	3.08	2.48	0.9400	3.81	2.91	4.22
		VCPA	9	13	0.9599	3.19	2.58	0.9445	3.76	3.06	4.28
		IRIV	**12**	**35**	**0.9748**	**2.54**	**2.03**	**0.9662**	**2.93**	**2.37**	**5.49**
		VCPA-IRIV	11	43	0.9808	2.21	1.71	0.9652	2.97	2.35	5.41
LLF	TVB-N	CARS	8	39	0.9457	2.42	1.89	0.9111	3.11	2.39	3.38
		VCPA	11	11	0.9303	2.75	2.09	0.9082	3.16	2.37	3.33
		IRIV	**14**	**76**	**0.9770**	**1.58**	**1.23**	**0.9431**	**2.49**	**1.95**	**4.23**
		VCPA-IRIV	12	62	0.9740	1.68	1.31	0.9426	2.49	1.98	4.21
	K value	CARS	9	45	0.9862	1.87	1.49	0.9614	3.14	2.43	5.13
		VCPA	11	11	0.9625	3.09	2.38	0.9466	3.69	2.89	4.36
		IRIV	14	94	0.9945	1.19	0.91	0.9752	2.51	1.95	6.40
		VCPA-IRIV	**14**	**73**	**0.9943**	**1.21**	**0.90**	**0.9815**	**2.17**	**1.63**	**7.40**

Notes: LVs, the latent variables.
VARs, The number of selected wavelength variables.
R^2C, Coefficient of determination of calibration.
R^2P, Coefficient of determination of prediction.
RMSEC, Root mean square error of calibration.
RMSEP, Root mean square error of prediction.
MAEc, Mean absolute error of calibration.
MAEp, Mean absolute error of prediction.
RPD, Residual predictive deviation of prediction.

relatively small. Machine learning techniques based on kernel function mapping or feature selection are particularly efficient at handling high-dimensional sparse data. In contrast, deep learning models typically require millions of samples to avoid overfitting, a phenomenon also observed in the study by Zhang et al. (Zhang et al., 2020). Despite the current limitations, the automatic feature learning ability of deep learning holds transformative potential as it circumvents the manual feature engineering required by PLS. In future research, a larger number of samples should be employed so that deep learning can be positioned as a forward-looking solution for intelligent systems.

3.5. External verification of experimental results

Validation of the model with an external dataset is a critical step to ensure the model's generalization ability and reliability. In this study, the same experimental time points and procedures as in Section 2 were adopted. Actual spectral data at each time point were collected, and for each time point, the TVB-N and K value were sampled three times to reduce experimental variability and enhance the reproducibility of the data. Subsequently, based on the LLF dataset, the optimal ML models, 2ndDer-IRIV-PLS and 2ndDer-VCPA-IRIV-PLS, as well as the optimal deep learning model, CNN-LSTM, were selected for validation experiments. The mean values of the three actual and predicted data points were calculated, and the error difference was computed. The experimental results are shown in Figure S3. The maximum errors for the traditional chemometric models were 2.32 mg/100 g and 2.05 %, while the maximum error for the deep learning model was 2.85 mg/100 g and 2.56 %, both of which fall within the anticipated error margins for the respective models. These results indicate that the predictive models developed in this study can effectively forecast the changes in TVB-N and K value during the spoilage process of shrimp flesh.

3.6. Visualization of the chemical information

Visualization, as a unique advantage of HSI, reveals the spatial distribution of substances, allowing a more intuitive understanding of the distribution and variation of TVB-N and K value during shrimp flesh degradation. This study employed the optimal IRIV and VCPA-IRIV models, based on Vis-NIR and NIR, to develop predictive models for TVB-N and K value. Visual maps of TVB-N and K value were constructed for each pixel, as shown in Fig. 4, representing the distribution of component content in shrimp flesh during five distinct stages of degradation. With increasing storage time, the maps demonstrate a transition in color from blue to red, indicating a clear shift in TVB-N and K value, and signify the shrimp flesh quality changes during refrigeration.

4. Conclusions

The study utilized HSI to analyze TVB-N and K value during shrimp flesh deterioration. By establishing ML and deep learning models, optimal prediction results were achieved on Vis-NIR, NIR and LLF data. The IRIV model based on LLF showed high prediction ability for TVB-N ($R_p^2 = 0.9431$, RMSEP = 2.49 mg/100 g, RPD = 4.23), while the VCPA-IRIV model based on LLF excelled in predicting K value ($R_p^2 = 0.9815$, RMSEP = 2.17 %, RPD = 7.40). Finally, the proposed machine learning model and deep learning model were compared and analyzed, which had important reference value for the research of shrimp products.

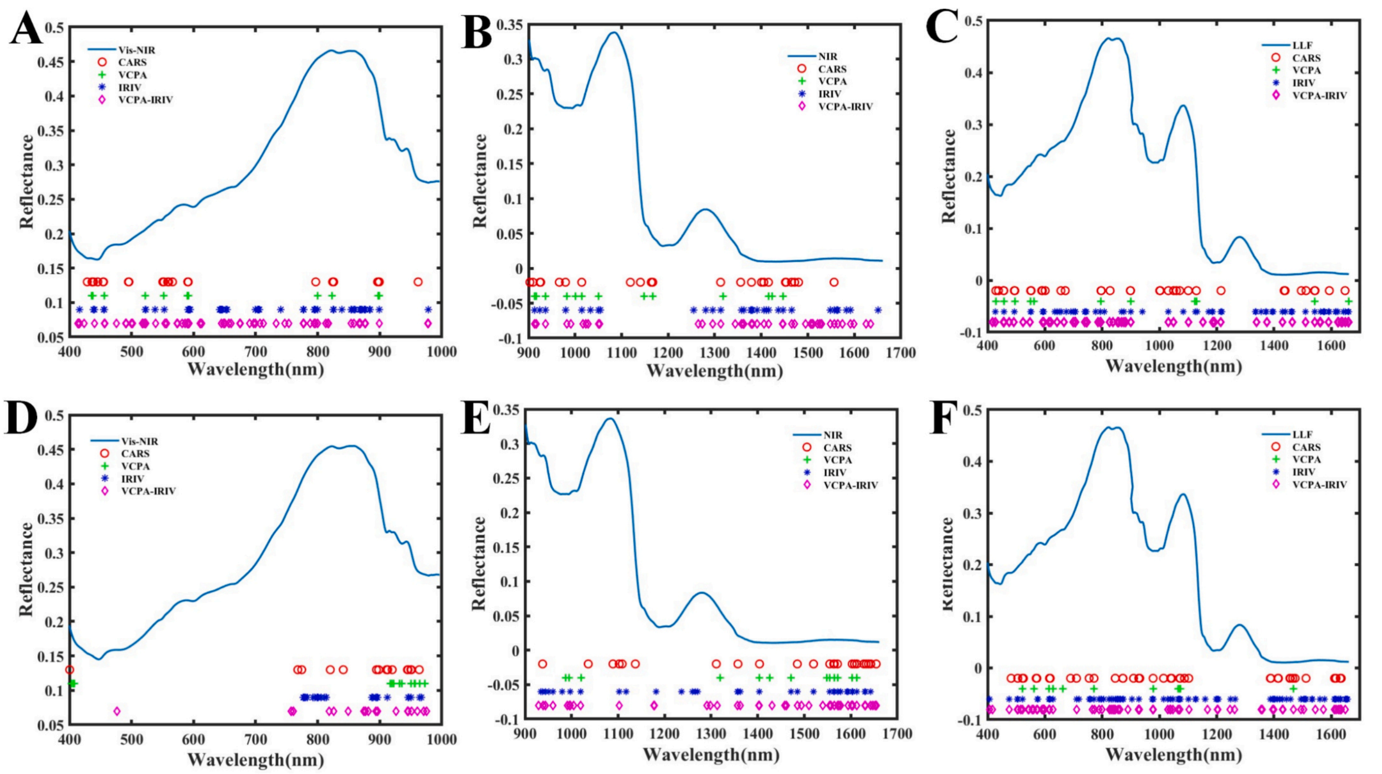

Fig. 3. The distribution of the optimal feature wavelengths in Vis-NIR, NIR, LLF. (A) Optimal features of TVB-N on Vis-NIR models, (B) optimal features of TVB-N on NIR models, (C) optimal features of TVB-N on LLF models, (D) optimal features of K value on NIR models, (E) optimal features of TVB-N on NIR models and (F) optimal features of K value on LLF models.

Table 3
Comparison of deep learning models based on Vis-NIR, NIR and LLF data.

Data	Compositions	Methods	Calibration			Prediction			RPD
			R^2c	RMSEC	MAEc	R^2p	RMSEP	MAEp	
Vis-NIR	TVB-N	CNN	0.9814	1.42	1.18	0.8369	4.21	3.25	2.54
		LSTM	0.9762	1.61	1.10	0.8528	4.00	3.00	2.63
		CNN-LSTM	**0.9467**	**2.40**	**1.73**	**0.8663**	**3.86**	**2.73**	**2.71**
	K value	CNN	0.9218	3.97	2.73	0.8954	4.59	3.26	3.10
		LSTM	0.9811	1.95	1.43	0.9034	4.42	2.74	3.24
		CNN-LSTM	**0.9851**	**1.74**	**1.24**	**0.9111**	**4.24**	**2.65**	**3.40**
NIR	TVB-N	CNN	0.8975	3.45	2.66	0.8551	4.09	3.22	2.64
		LSTM	0.8558	4.21	3.13	0.8158	3.98	3.22	2.36
		CNN-LSTM	**0.8968**	**3.46**	**2.44**	**0.8682**	**3.89**	**3.21**	**2.92**
	K value	CNN	0.9113	4.75	3.71	0.8808	5.51	4.28	2.92
		LSTM	0.9314	4.18	3.25	0.8872	5.37	3.97	3.00
		CNN-LSTM	**0.9699**	**3.85**	**1.15**	**0.9033**	**4.96**	**3.59**	**3.24**
LLF	TVB-N	CNN	0.9436	3.13	2.22	0.8901	3.45	2.39	3.02
		LSTM	0.9426	3.32	2.34	0.9140	3.05	2.37	3.06
		CNN-LSTM	**0.9630**	**2.38**	**1.29**	**0.9273**	**2.99**	**2.30**	**3.48**
	K value	CNN	0.9699	2.94	2.26	0.9115	4.75	3.71	3.39
		LSTM	0.9719	2.68	2.05	0.9259	4.34	3.25	3.77
		CNN-LSTM	**0.9895**	**1.66**	**1.29**	**0.9675**	**2.88**	**1.83**	**5.59**

Notes: R^2_C, Coefficient of determination of calibration.
R^2_P, Coefficient of determination of prediction.
RMSEC, Root mean square error of calibration.
RMSEP, Root mean square error of prediction.
MAEc, Mean absolute error of calibration.
MAEp, Mean absolute error of prediction.
RPD, Residual predictive deviation of prediction.

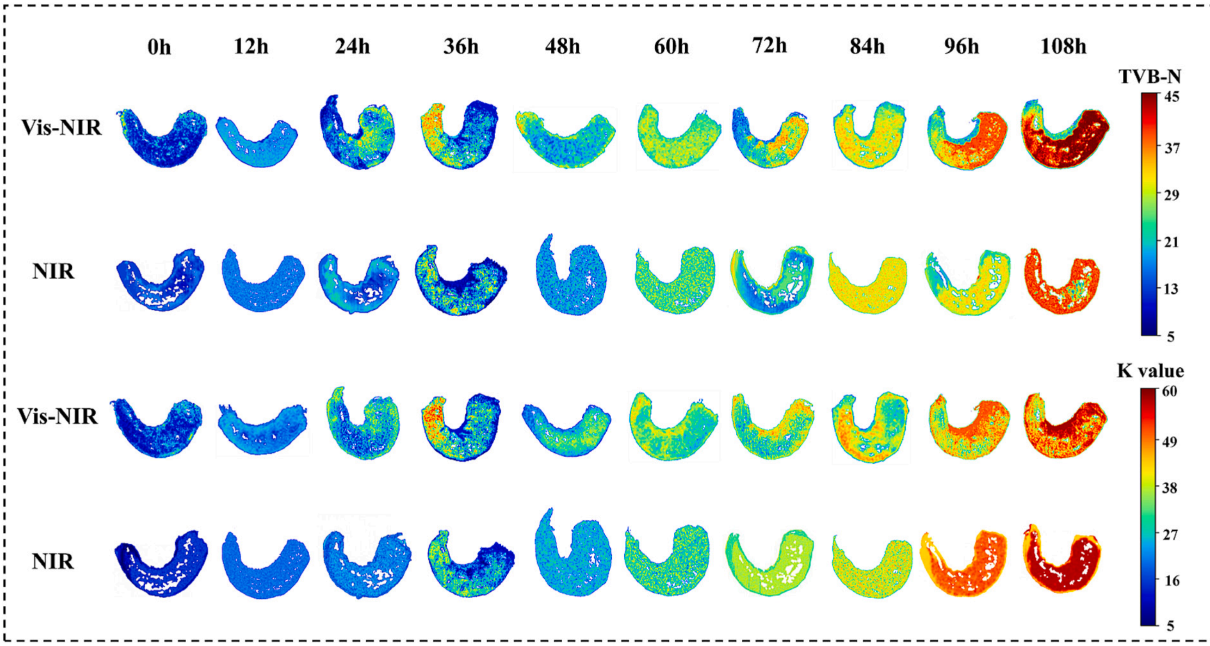

Fig. 4. Visualization of TVB-N and K value using Vis-NIR and NIR at ten time points.

Despite these promising results, several limitations should be considered. Due to the relatively small sample size in this study, the ability to apply the findings to a broader population may be limited. Additional challenges include the experimental environment, the use of a single species, and the high cost of equipment, all of which could impose constraints on the future progress of this study. Future research will aim to overcome these limitations and enhance the application of HSI technology in monitoring food quality.

CRediT authorship contribution statement

Qibing Xi: Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Qingmin Chen:** Writing – review & editing, Validation. **Waqas Ahmad:** Writing – review & editing. **Jing Pan:** Investigation, Formal analysis. **Songguang Zhao:** Methodology, Formal analysis. **Yu Xia:** Investigation. **Qin Ouyang:** Project administration, Funding acquisition, Data curation. **Quansheng Chen:** Supervision, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foodchem.2025.143997.

Data availability

Data will be made available on request.

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